Pratik Nanaso Raut

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PROFESSIONAL SUMMARY

Computer Engineering candidate with strong foundation in Java, Spring Boot, REST APIs — backend development.

TECHNICAL SKILLS

Languages	Java, SQL, C, C++, JavaScript, Python (basics)
Backend	Spring Boot, REST APIs, Spring Security, Hibernate, JWT, JDBC
Databases	MySQL, MongoDB
Tools	Git, GitHub, Maven, Postman, IntelliJ IDEA
Concepts	SDLC, Object-Oriented Programming (OOP), Data Structures & Algorithms (DSA), Design Patterns, REST API Design, Problem Solving, Time Complexity Analysis, Database Optimization, API Security, RBAC, JWT Authentication, Microservice Fundamentals, Multi-Tenant Architecture, Multithreading

PROFESSIONAL EXPERIENCE

Software Developer Intern *Feb 2026 -- Ongoing*
SetTribe IT Solutions | Pune

- Developed and enhanced ERP backend modules using Java, Spring Boot, Hibernate, and REST APIs for employee and operational workflows.
- Integrated PostgreSQL database operations, validation logic, and API testing workflows using Postman.
- Collaborated in Agile sprint cycles, performing Git-based code reviews, bug fixes, feature implementation, and deployment support.

ACADEMIC PROJECTS

Customised Virtual File System (CVFS) -- In-Memory Linux-like File Manager *C, C++, Data Structures, OS Concepts*

- Simulated Linux file system operations in C++ using inode-based metadata.
- Achieved O(1) file lookup by designing hash table-based inode indexing.
- Implemented in-memory file allocation table managing file permissions.

ProcMemInspector -- Linux Process Memory Layout Analyzer *C, Linux Internals, /proc Filesystem*

- Developed /proc filesystem parser to display virtual memory layout.
- Implemented process statistics monitoring by reading /proc/[pid]/maps.
- Analyzed virtual address space segmentation demonstrating memory isolation.

Concurrent FTP Server -- Multi-Client File Transfer Protocol Server *C, POSIX Threads, TCP Sockets, Multithreading*

- Engineered concurrent client handling using POSIX threads over TCP sockets.
- Implemented thread-safe resource management using mutex and semaphore primitives.
- Designed application-layer protocol over raw sockets.

TECHNICAL STRENGTHS

- Proficient in backend application development, REST API design, and database-driven systems.
- Strong foundation in C++ OOP, STL containers, templates, and dynamic memory management.
- Proficient in POSIX system calls, concurrent programming, and Linux kernel interface design.
- Practical exposure to network programming, socket communication, and concurrent server develop-

ment.

EDUCATION

Master of Computer Applications (MCA) *2026 | Pursuing*
Dnyansagar Institute of Management and Research, Pune (SPPU) 7.24 CGPA **Bachelor of Computer Applications (BCA)** *2024*
C.T. Bora College, Shirur (SPPU) 73.34% **Higher Secondary Certificate (HSC)** *2021*
Maharashtra State Board 75%